Name ______________________________ Section ________

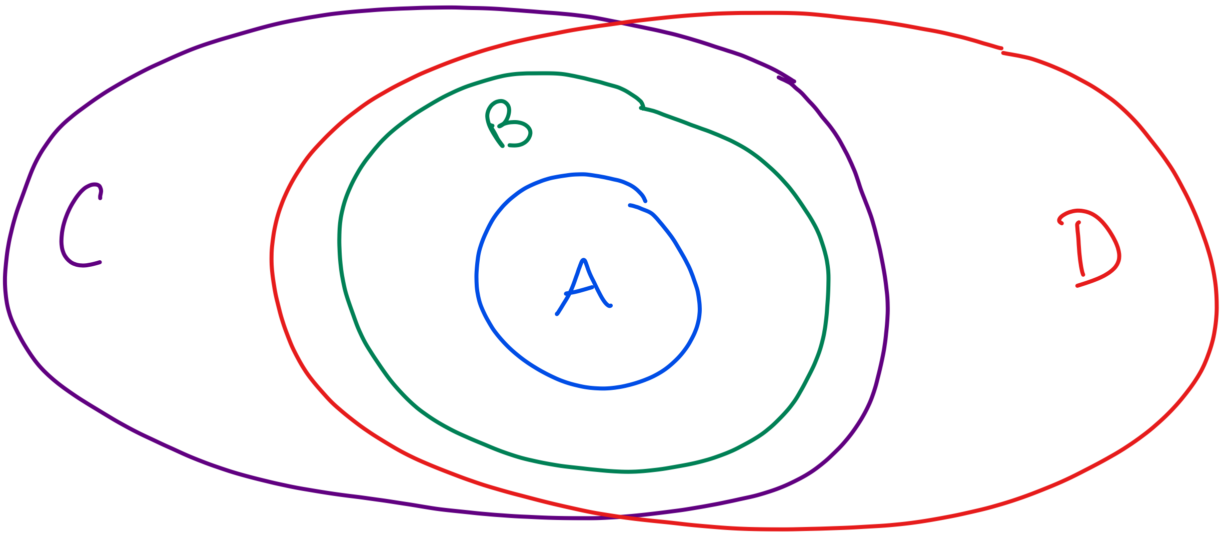
MTH:2310 DISCRETE MATH
QUIZ 5
DUE: MONDAY MAR. 10

Complete all of the following problems being as detailed as you can be. You may work in groups, but you must submit your own solutions using your own words. You must submit a physical copy of your solutions to me unless told otherwise. If there is a specific number of problems which seem unsolvable let me know first, I may have made a typo. Even if you cannot completely solve a problem, a good description of key terms and concepts relevant to the problem is sufficient for some partial credit.

Handwriting: (4 points)
Your ability to express your solutions is just as important as mathematical accuracy.
You will be penalized up to 2 points for arithmetic errors.
You will be penalized up to 2 points for disorganized or hard to read solutions.

Problem 1. *(12 points) Set Operations*
Let U be the universal set. Let A, B, C and D be distinct non-empty sets. Suppose $A \subseteq B$, $A \subseteq C$, $B \subseteq C$, and $B \subseteq D$.

(a) *(8 points)* Use a single Venn diagram to illustrate all of these relationships.



(b) *(2 points)* Is it necessary that $A \subseteq D$? Briefly justify why or why not.

Yes $A \subseteq B$ & $B \subseteq D$
so $A \subseteq D$

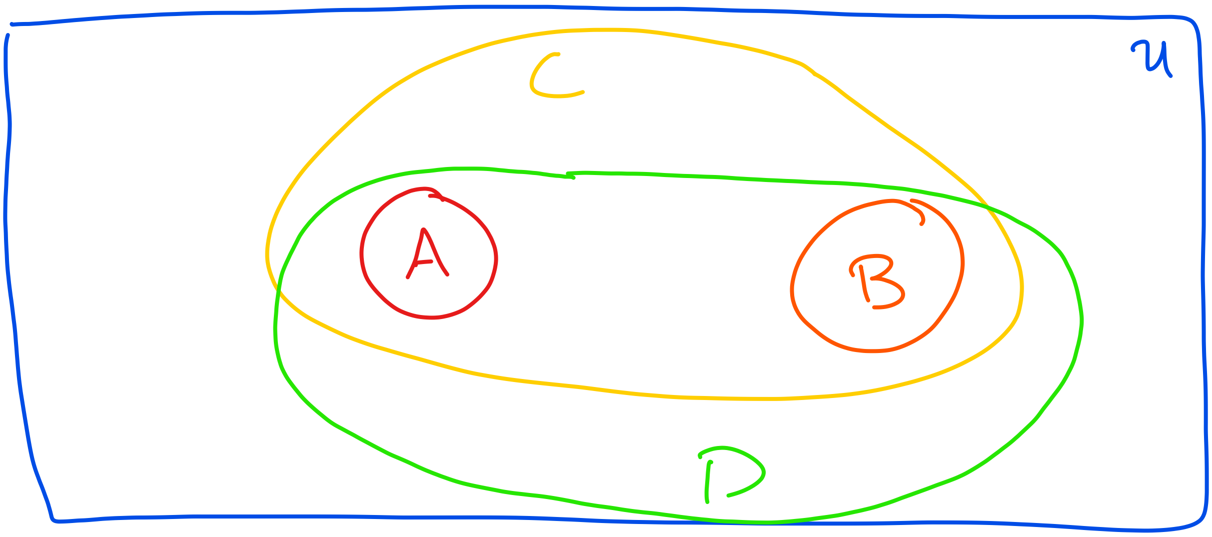
(c) *(2 points)* Is it necessary that $C \subseteq D$? Briefly justify why or why not.

No, $C - D$ can be non-empty and still satisfy the properties. See (a)

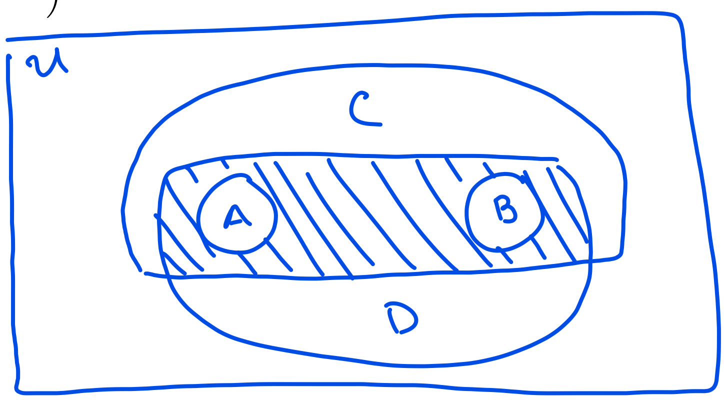
Problem 2. *(22 points) Set Operations*
Let U be the universal set. Let A, B, C, and D be distinct non-empty sets. Suppose that

- $A \subseteq C$
- $B \subseteq C$
- $C \not\subseteq D$
- $A \cap B = \emptyset$
- $A \subseteq D$
- $B \subseteq D$
- $D \not\subseteq C$

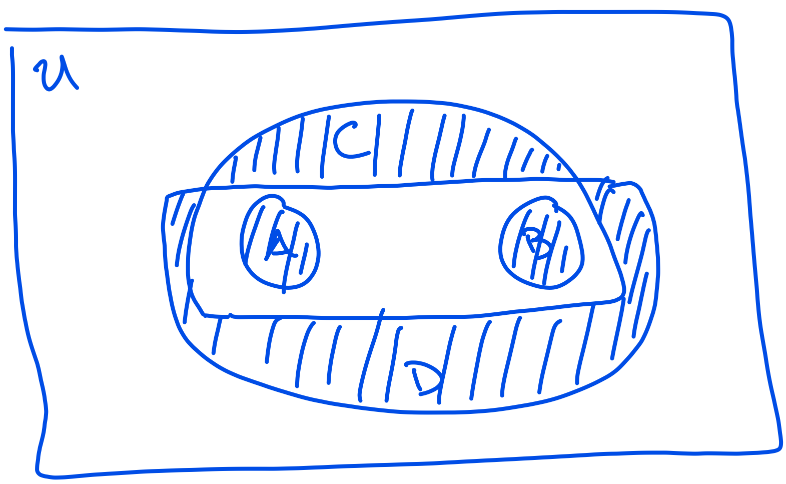
(a) *(14 points)* Use a Venn diagram to illustrate all of these relationships.



(b) *(4 points)* Redraw the diagram and shade the region corresponding to $\overline{A \cup B} \cap \left(C \cap D\right)$



(c) *(4 points)* Redraw the diagram and shade the region corresponding to $\left(\left(C \cup D\right) \setminus \left(C \cap D\right)\right) \cup \left(A \cup B\right)$



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Problem 3. *(8 points) Set Operations*
Let A, B, and C be non-empty sets. Show that $\big(B \setminus A\big) \cup \big(C \setminus A\big) = \big(B \cup C\big) \setminus A$ using set-builder notation only.

$$(B \setminus A) \cup (C \setminus A) = \{x \mid (x \in B \setminus A) \vee (x \in C \setminus A)\}$$

$$= \{x \mid ((x \in B) \wedge (x \notin A)) \vee ((x \in C) \wedge (x \notin A))\}$$

"Reverse" Distributive Property

$$= \{x \mid ((x \in B) \vee (x \in C)) \wedge (x \notin A)\}$$

$$= \{x \mid (x \in B \cup C) \wedge (x \notin A)\}$$

$$= (B \cup C) \setminus A$$

Problem 4. *(8 points) Set Operations*
Let A, B, and C be non-empty sets and U be the universal set. Show that $A \cap \overline{B \cup C} = \big(A \setminus B\big) \setminus C$ using set-builder notation only.

$$A \cap \overline{B \cup C} = \{x \mid (x \in A) \wedge (x \in \overline{B \cup C})\}$$

$$= \{x \mid (x \in A) \wedge ((x \in U) \wedge (x \notin B \cup C))\}$$

$$= \{x \mid (x \in A) \wedge ((x \in U) \wedge \sim(x \in B \cup C))\}$$

$$= \{x \mid (x \in A) \wedge ((x \in U) \wedge \sim((x \in B) \vee (x \in C)))\}$$

$$= \{x \mid (x \in A) \wedge ((x \in U) \wedge (x \notin B) \wedge (x \notin C))\}$$

$$= \{x \mid \underbrace{(x \in A) \wedge (x \in U)}_{\equiv\, x \in A} \wedge (x \notin B) \wedge (x \notin C)\}$$

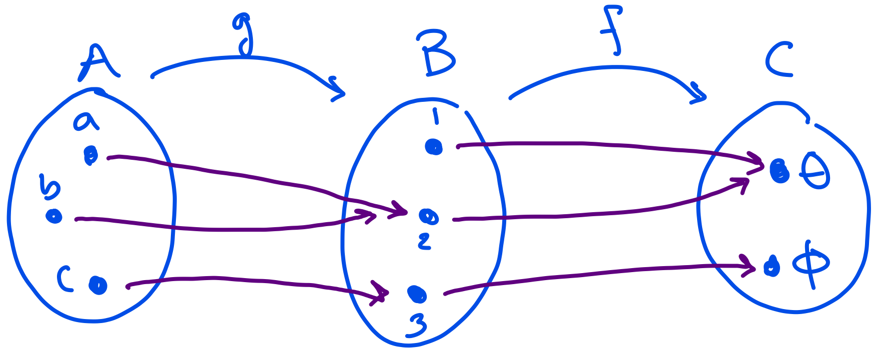
$$= \{x \mid (x \in A) \wedge (x \notin B) \wedge (x \notin C)\}$$

$$= \{x \mid (x \in A \setminus B) \wedge (x \notin C)\}$$

$$= (A \setminus B) \setminus C$$

Problem 5. *(7 points) Functions*
Let $f : B \to C$ and $g : A \to B$ be functions where $A = \{a, b, c\}$, $B = \{1, 2, 3\}$, and $C = \{\theta, \phi\}$. Define f and g in such a way that the composition $f \circ g$ is surjective, but g is not surjective. You may use an explicit definition or a "bubble-arrow" diagram.

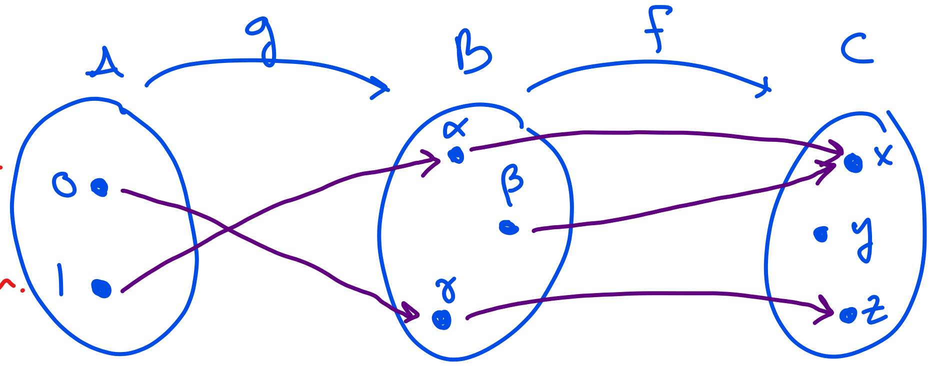
g is ~~not~~ not sur. since Range(g) does not include 1.



$f \circ g$ is sur.
$(f \circ g)(a) = \theta$
$(f \circ g)(c) = \phi$

Problem 6. *(7 points) Functions*
Let $f : B \to C$ and $g : A \to B$ be functions where $A = \{0, 1\}$, $B = \{\alpha, \beta, \gamma\}$, and $C = \{x, y, z\}$. Define f and g in such a way that the composition $f \circ g$ is injective, but f is not injective. You may use an explicit definition or a "bubble-arrow" diagram.

$f \circ g$ is inj. since $(f \circ g)(\ ) = x$ has only 1 solution and $(f \circ g)(\ ) = z$ has only 1 solution.



f is not inj.
$f(\alpha) = f(\beta)$

Problem 7. *(8 points) Functions*
Prove that the function $f : \mathbb{R} \to \mathbb{R}$ is injective where f is defined as $f(x) = x^3$

Let a, b be two arbitrary real numbers.
Assume $f(a) = f(b)$. Then $a^3 = b^3$.
The only real solution for this is when
$a = b$.
Therefore $f(x) = x^3$ is injective.